

PhD MILESTONE 1 REPORT



Economic Model Predictive Control of a Semi-Autogenous Grinding Mill Circuit: A Modular Energy-Minimising Cost Function

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August 2026

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1. Abstract

Semi-autogenous grinding (SAG) is the most energy-intensive stage of mineral comminution. The comminution circuit dominates a concentrator’s electricity bill and the greenhouse emissions that come with it. The control objective is therefore economic. The mill should grind the ore to the size the downstream circuits require, at the lowest possible energy per tonne, while respecting throughput demand and product-quality limits. The classical solution places a steady-state real-time optimiser (SSRTO) above a tracking model predictive controller (MPC). That two-layer design is poorly matched to a SAG circuit. Ore hardness and feed size drift on hour-to-shift timescales, so the plant rarely holds the strict steady state that SSRTO’s parameter-estimation step requires. The optimiser then sits idle and pushes stale set-points. Economic MPC (EMPC) merges optimisation and regulation into a single receding-horizon problem. It can track a moving economic optimum during transients, and transients are where a SAG mill spends most of its operating life.

This research develops and benchmarks an EMPC framework for a SAG circuit. It uses two prediction models. The Hulbert-Le Roux reduced-order model handles fast online solves. A cumulative-rates population balance model (PBM) covers cases that need the full product size distribution. A fast disturbance observer (DO) runs at the regulatory clock. An augmented extended Kalman filter (EKF) tracks the slow drift in ore hardness at the economic clock. Two contributions are central. The first is a modular, energy-minimising cost function. The online controller minimises mill energy alone. The downstream economics do not enter it directly. They are worked out offline and reduced to one specification, a particle-size floor the controller must hold. Re-tuning for a new ore body or a different downstream plant then means swapping that specification, not redesigning the controller. The second is a comparison of the supervisory architectures (SSRTO, ROPA, and single- and two-layer EMPC) against a conventional PI / tracking-MPC baseline. The test is not a fixed steady state but a plant-representative, transient-rich scenario. This matters because the preliminary results show the architectures behave almost the same at a single operating point. They only differ once the plant has to move between operating points.

Preliminary work has implemented all of these controllers in `do-mpc`/CasADi/IPOPT on a simulated eight-state circuit. They have been exercised under drifting disturbances, and each behaves as theory predicts. These first tests are synthetic and largely confined to fixed set-points. On them the supervisory architectures return broadly comparable results. They therefore neither license plant-level numbers nor separate the architectures from one another. The substance of the comparison lies ahead. The mill model must be calibrated to historical plant data. A realistic synthetic scenario must be built, rich in the transients and disturbances a SAG circuit actually experiences. That scenario then measures the real differences between the architectures in an apple-to-apple comparison that varies only the supervisory layer.

Nomenclature

Manipulated variables (MV): MFS mill feed solids / ore feed rate (t/h) · CFF cyclone feed flow · SFW sump feed water · c mill speed (fraction of critical speed, 0.55-0.80).

Controlled variables (CV): JT total mill charge filling · SVOL sump slurry volume · PSE product size estimate (fraction < 75 μm) · Pmill mill power draw.

Plant states (8): mill X_{mw} , X_{ms} , X_{mf} , X_{mr} , X_{mb} (water, solids, fines, rocks, balls); sump X_{sw} , X_{ss} , X_{sf} (water, solids, fines).

Disturbances / parameters: r feed rocks fraction · f ore-hardness (power per tonne of fines) · MIW mill inlet water · MFB ball feed.

Derived / economic: TP throughput (MFS at steady state) · SEC = Pmill/MFS specific energy consumption (kWh/t) · A net return per tonne (\$/t) · B electricity tariff (\$/kWh) · NSR net smelter return · J_comm commercial objective.

Acronyms: SAG semi-autogenous grinding · PBM population balance model · PSD particle-size distribution · (E)MPC (economic) model predictive control · RTO real-time optimisation · SSRTO steady-state RTO · ROPA RTO with persistent adaptation · DRTO dynamic RTO · EKF extended Kalman filter · DO/NDOB (nonlinear) disturbance observer · PINN physics-informed neural network · OCP optimal control problem · NLP nonlinear program.

2. Research Background

2.1 The SAG grinding circuit and its control problem

Comminution is the crushing and grinding of run-of-mine (ROM) ore to free the valuable mineral. It is the single largest electricity consumer on a concentrator, and the SAG mill is the largest single draw within it. The control problem is multivariable and strongly coupled. A closed circuit (SAG mill $\rightarrow$ sump $\rightarrow$ hydrocyclone) is driven through a handful of manipulated variables: ore feed rate, cyclone feed flow, dilution water and mill speed. The controller must hold a few controlled variables (mill charge, sump level, product particle size) inside an operating envelope. Meanwhile the ore properties drift underneath the controller. The economically meaningful quantity is energy per tonne of product at acceptable quality. The natural objective is therefore not to track a set-point but to optimise the energy-throughput-quality trade-off directly. This motivates economic, rather than purely regulatory, control.

A controller is only as good as the model behind it, the estimator that feeds it state, and the supervisory layer that decides *where* to operate.

2.2 Control-oriented dynamic models

A control-oriented mill model must be predictive enough to capture the load, power, and product dynamics. It must also stay cheap enough to solve inside an online optimisation. Two mechanistic families span this trade-off, and both are used in this work. A reduced-order lumped model serves the fast economic MPC. A size-resolved population balance model serves the cases where the cost function must see the full product size distribution.

The **Hulbert-Le Roux reduced-order model** (Le Roux et al., 2013; Le Roux & Steyn, 2022) collapses the size structure to three operationally meaningful lumps: *rocks*, *solids* and *finer*. Water and ball hold-ups are added, giving a five-state mill plus a three-state sump in the closed-circuit form. Its strengths make it the default for online control. It can be identified from a single plant survey. It supports non-linear MPC at industrial sample rates (Coetzee et al., 2010; Le Roux et al., 2016). It admits a clean EKF observer (Le Roux et al., 2017) and tolerates mode-switching between start-up and near-overload regimes (Botha et al., 2018). It has also been embedded in an industrial digital twin (Quintanilla et al., 2025). Its drawbacks follow from the same lumping. The single quality output is the product particle-size estimate (PSE), the fraction of cyclone-overflow product passing a specified limit (e.g. 75 μm). The model therefore represents only one quantile of the size distribution and never its shape. Its empirical breakage and discharge functions are also fitted to a particular circuit.

The **cumulative-rates population balance model** resolves the full PSD. It tracks the ore mass in each of n geometrically-spaced size classes, giving a $2n+2$ -state circuit (n mill classes plus mill water, n sump classes plus sump water).

Its defining feature is the cumulative-rates parametrisation of Hinde & Kalala (2009). A standard PBM describes breakage with two functions that are hard to measure: a specific breakage-rate function and a breakage-distribution function. Their parameters are back-calculated from test data and are notoriously sensitive to measurement error, with large variances. The cumulative-rates form replaces both with a single cumulative breakage rate $K_i \hat{E}$. This is the rate at which material coarser than size x_i breaks below it, normalised by mill power so that it is largely insensitive to scale-up. Its decisive advantage is direct estimation. $K_i \hat{E}$ comes straight from measured feed and product size distributions and the mill’s specific energy input, with no back-calculation. That makes a PSD-resolving model robustly identifiable, and smooth enough to linearise or differentiate for control.

The two families are complementary rather than competing. The Hulbert-Le Roux model is preferred for online MPC. Its small state keeps the optimisation cheap and offers the right granularity for grinding-circuit-only (PSE-based) economic decisions. The cumulative-rates PBM is needed when the cost function must see PSD shape, for example when downstream flotation and leaching kinetics are size-dependent. Any genuinely plant-wide economic objective requires a PSD-resolving signal that a single PSE quantile cannot provide. The reduced $n = 5$ or 9 form supplies that at a state count a controller can still afford.

2.3 Disturbance observers and state estimation

A SAG circuit’s disturbances are well characterised and act on distinct timescales. Ore-hardness drift acts hour-to-shift and is the dominant source of model-plant mismatch. Feed-size variability acts on minutes-to-hours and is usually correlated with hardness. Feed-moisture and dilution-water swings change the slurry rheology. Liner wear and ball-charge drift act over weeks to months. The literature splits the estimation work across two roles. A fast non-linear disturbance observer (NDOB) at the regulatory MPC layer cancels input-side disturbances by feed-forward before they reach the optimisation loop. The first grinding-circuit deployments (Chen et al., 2009; Yang et al., 2010) established two things. Pairing a DO with the MPC improves disturbance rejection while leaving nominal tracking intact. The estimate also stays meaningful to operators, who can read it directly as an “ore-hardness offset”. The second role is a slow estimator at the economic layer, an augmented EKF/UKF (Le Roux et al., 2017). It tracks the drifting hardness parameter and recovers the latent mill states. The augmented Kalman filter is the established choice for this role on grinding circuits. It propagates the slow parameter drift through its process-noise model while staying a deterministic, real-time update. Here the augmented estimator tracks a single slow parameter, the ore hardness $\mathbf{f}$ (the power drawn per tonne of fines produced). One direction this work will explore is augmenting it with further slowly-varying disturbance states, such as the feed rocks fraction $\mathbf{r}$ and other rheology-related parameters. The test is whether jointly estimating them, rather than $\mathbf{f}$ alone, sharpens the

economic-layer model under the realistic, multi-disturbance scenario.

2.4 The supervisory-optimisation continuum: SSRTO $\rightarrow$ ROPA $\rightarrow$ DRTO $\rightarrow$ EMPC

The question “*how does the controller decide what the economically best operating point is?*” has four published answers. **Classical SSRTO** (Darby et al., 2011) puts a separate steady-state optimiser above the MPC. It fits a steady-state model to plant data collected while the plant is at steady state, then solves a static economic optimisation for the next set-points. Fitting the model to transient data corrupts it, so the optimiser has to wait for the plant to settle, and each cycle ends up taking hours. For a SAG mill this is a real problem. The plant rarely stays at steady state for long, so the optimiser runs seldom and works from stale data.

ROPA (real-time optimisation with persistent adaptation) breaks SSRTO’s central coupling. It runs a continuous dynamic estimator for the model parameters, so the still-steady-state economic NLP always receives fresh parameters, even during transients. **DRTO** replaces the static NLP with a dynamic optimisation over a long horizon. It emits an economic *trajectory* for a tracking MPC to follow. **EMPC** (Ellis et al., 2014; Faulwasser & Pannocchia, 2018) takes the DRTO logic to its limit. It drops the dedicated upper layer and embeds the economic objective directly in the regulatory MPC’s stage cost. DRTO and EMPC are theoretically equivalent. They share the same dynamic economic cost, model and horizon, and differ only in whether the layers are kept visibly separate. An experimental-rig comparison (Matias et al., 2022) reports ROPA matching DRTO at about 38 % profit improvement over a fixed-input baseline, while SSRTO reaches only about 20 %. The practical ordering is EMPC > DRTO > ROPA >> SSRTO.

2.5 EMPC on grinding circuits - the empirical record

The components have been published piecewise. Thivierge, Bouchard and Desbiens (2023) provide the closest precedent. They compare EMPC against basic and advanced regulatory control on a simulated HPGR-ball-mill-flotation circuit, with a cost function of flotation-concentrate revenue minus HPGR specific energy. However, they evaluate only at steady-state operating points after the simulator re-settles, and defer transient performance to future work. Numbi & Xia (2016) demonstrate energy-cost EMPC on a crushing plant under time-of-use tariffs. Jia et al. (2020) demonstrate multi-stage EMPC for downstream gold cyanidation. No single study combines the Hulbert-Le Roux model, a composite DO + augmented-EKF estimator stack, and an explicit specific-energy EMPC cost on a SAG circuit, evaluated under realistic operating conditions.

3. Research Gap and Problem Statement

The literature reviewed in Section 2 traces a body of work in which every component of an EMPC for a SAG circuit has been published. They have never been assembled into a single closed-loop study. They have also never been exercised under the conditions an industrial SAG mill actually experiences. Three connected gaps follow.

Gap 1 - Architectural integration on a SAG circuit. No published study integrates a regulatory-layer DO, an upper-layer augmented EKF/UKF, and an economic stage cost into a single closed-loop SAG-mill controller. The two estimator strands address complementary timescales but have evolved independently. DO + MPC cancels fast input-side disturbances at the regulatory clock. The augmented-KF strand absorbs slow parameter drift at the economic clock. On a SAG mill these timescales coexist. A controller built around either layer alone therefore leaks performance through the timescale it does not handle.

Gap 2 - Realistic operating conditions in evaluation. Industrial SAG mills settle at *operational* steady state often, but the underlying conditions never do. Ore hardness drifts hour-to-shift. Liner wear changes the mill geometry over weeks to months. The plant therefore moves between mildly suboptimal operating points rather than running at the economic optimum. Published grinding-circuit studies report performance under tightly controlled conditions instead: step disturbances at fixed operating points, with metrics sampled after re-settling. Two things have not been quantified. The first is how a SAG-mill EMPC behaves under disturbance sequences that resemble actual plant operation. The second is how large the economic opportunity is that EMPC could capture by tracking the moving optimum during transients. This quantification is the prerequisite for any fair head-to-head comparison of EMPC against SSRTO, ROPA, or DRTO.

Gap 3 - Cost-function scope on a SAG circuit. The plant-wide framework of Le Roux & Craig (2019) and the flotation-revenue cost of Thivierge (2023) both argue that the mill’s economic objective should pull from downstream plant signals. Most grinding-circuit EMPC studies accordingly adopt a revenue- or profit-based stage cost. Two problems with that choice on a single-stage SAG mill have gone unaddressed. The first is a scale mismatch. A profit term such as $A(PSE) \cdot TP - B \cdot P_{mill}$ prices the product through a per-tonne return $A(PSE)$. That one number collapses recovery, grade, metal price and downstream constraints. All of these are properties of the *plant*, set by the flotation and leaching state and by market conditions the mill controller neither measures nor manipulates. A mill-level EMPC carrying that term must reconstruct an online proxy for the entire downstream plant from its one quality signal. The proxy is blind to the recovery and grade actually realised downstream, so the mill’s local optimum can diverge from the true plant-wide optimum. The second problem is a trivial optimum. Even with A known, revenue scales with throughput while energy is a small per-tonne penalty. Maximising the profit form therefore simply

drives mill speed ϕ_c and feed MFS to their limits. The mill runs flat out against its throughput and overload constraints. That is a maximum mode that needs no optimiser, not a meaningful interior optimum.

A complementary and unexplored direction resolves both problems: make the cost function modular and mill-local. The throughput demand (TP/MFS) and the product-quality floor (PSE) are taken as given. They are set by plant-wide coordination, which balances the circuit’s units rather than running each flat out. This turns the degenerate maximum-mode chase into a genuine minimisation over the freedom that remains. The stage cost is then composed from a small, extensible library of mill-owned terms. The library holds mill energy ($B \cdot P_{\text{mill}}$), specific energy consumption (SEC, energy per tonne of feed ore), and penalties on mill-specific quantities such as residence time. With a size-resolved PBM it can also penalise the coarse tail of the product PSD. Terms are used singly or in combination, and weighted, as the downstream specification requires. Both targets reach the controller only through a thin, offline-derived specification interface. The interface has two variables. The first is the throughput demand MFS/TP, set by the feeders, crushers and production plan. The second is a downstream product-quality floor (PSE) with a frozen per-tonne return that summarises the recovery economics. The mill minimises energy per tonne within that interface. It never tries to recover profit by chasing throughput itself. Every ore-, plant-, and commodity-specific quantity lives in the interface rather than in the cost function. The same SAG-mill controller therefore transfers to a new ore body, a different downstream circuit, or another commodity, by re-deriving the interface offline. Neither the modular framing nor any of these mill-local objectives has yet been instantiated as the closed-loop cost function of a single-stage SAG mill.

These three gaps define the problem this thesis addresses: to build an integrated single-stage SAG-mill EMPC with a composite DO + augmented-estimator stack, to give it a modular energy-minimising cost function, and to benchmark it honestly against the full RTO/EMPC architecture spectrum under operating conditions representative of a real plant.

4. Aims and Objectives

The aim of this research is to develop and rigorously evaluate an economic model predictive control framework for a single-stage SAG grinding circuit that minimises specific energy consumption under realistic operating conditions, and to establish where it offers a genuine advantage over the established real-time-optimisation architectures. This aim is pursued through four objectives.

1. **Implement and validate the integrated control stack.** Build the single-stage SAG controller in `do-mpc`/CasADi/IPOPT and verify closed-loop regulation against the simulated plant. The controller runs on both the Hulbert-Le Roux reduced-order model and the cumulative-rates PBM,

used in parallel according to the strengths and drawbacks of each (Section 2.2). Estimation uses the EKF / augmented EKF plus a fast disturbance observer. (*Substantially complete.*)

2. **Formulate a modular, energy-minimising EMPC cost function.** Derive and implement a stage cost in which the downstream plant economics enter only through an offline-derived particle-size specification (a one-sided PSE/PSD floor). The online controller then minimises mill energy at the demanded throughput and quality. Demonstrate the modularity claim: adapting to a new ore body or downstream topology is a swap of the specification interface, not a controller redesign (the interface is detailed in Section 6.2). Explore modular cost functions suited to both single- and multi-stage SAG circuits, where each mill's local objective couples to the next only through the specification interface. (*Primary contribution.*)
3. **Construct a realistic transient scenario and benchmark the architectures.** Build a plant-representative, transient-rich operating scenario calibrated to historical plant data. It combines ore-hardness drift, feed-size variability, and operating-point changes that drive the plant between steady states. Use it to compare SSRTO, ROPA, and single- and two-layer EMPC against a decentralised-PI / tracking-MPC baseline, on specific energy consumption and constraint compliance. (*Primary contribution.*)
4. **Develop a PINN surrogate for online tractability.** Train a physics-informed neural-network surrogate offline on PBM-simulated data. Substitute it for the mechanistic prediction model inside the EMPC. The goal is to keep the online optimisation tractable as model fidelity and horizon length grow.

Scope and priority. Objectives 1-3 form the core of the candidature and are largely instantiated in the preliminary work. **Objectives 2 and 3 are the central novel contributions** (the modular cost function and the transient benchmark). Objective 4 (PINN) is optional. It is pursued only if online solves become too slow as the horizon and model detail grow. Otherwise the mechanistic model stays the default. This staging keeps the thesis deliverable even if Objective 4 yields negative or partial results.

5. Significance and Novelty

5.1 Theoretical contributions

This work would be the first to bring three pieces together inside a single closed-loop SAG-mill controller: a regulatory-layer disturbance observer, an upper-layer augmented EKF, and an explicit specific-energy economic stage cost. That closes the architectural gap the component literature leaves open (Gap 1). The modular spec-interface cost function (Objective 2) is a novel framing of the plant-wide economic objective. It separates the offline downstream-economics question from the online control problem. The transient-scenario benchmark (Objective 3) provides the first quantification on a SAG mill of the economic opportunity EMPC captures during transients. That measurement is what any honest EMPC-vs-RTO comparison requires.

5.2 Practical contributions

The outcome is a portable, energy-minimising SAG-mill controller. Its economic objective can be retargeted to a different ore body or concentrator configuration by replacing an offline-derived specification, not by re-engineering the controller. That lowers the barrier to industrial reuse. The transient-scenario study yields credible, plant-representative specific-energy-saving figures rather than figures from a benign synthetic test. Practitioners gain a defensible basis for deciding between an RTO retrofit and a full EMPC. Comminution dominates concentrator electricity use; grinding alone can draw up to roughly 80 % of a concentrator’s energy. Even single-digit-percent specific-energy reductions therefore translate into material cost and emissions savings at industrial scale. By cutting grinding electricity demand and the associated greenhouse emissions, the project supports Western Australia’s Net Zero by 2050 commitment. It also aligns with Curtin University’s research direction in sustainable mineral processing.

6. Research Methodology

The methodology is organised in three parts: the process under study and its control problem (Section 6.1), the control architecture proposed for it (Section 6.2), and the forward work plan that delivers and evaluates the contributions phase by phase against the objectives of Section 4 (Section 6.3).

6.1 The single-stage SAG grinding mill circuit

The case study throughout is a single-stage grinding mill circuit closed by a hydrocyclone (Figure 1). This is the configuration most commonly used for single-stage grinding and the standard testbed for grinding-circuit control (le Roux et al., 2013). Its three elements are a SAG mill, a sump, and a cyclone. A SAG mill is chosen for three reasons. It has a low operating cost relative to

conventional grinding. It processes large tonnages of low-grade ore at reduced grinding-media consumption. It also accepts coarse run-of-mine feed that would otherwise require additional crushing.

Run-of-mine ore (MFS), steel balls (MFB), and inlet water (MIW) are fed to the mill. The mill is driven at a fraction of critical speed (ϕ_c). It grinds the charge and discharges slurry to the sump. Dilution water (SFW) sets the density there, and the cyclone feed pump delivers the slurry at flow CFF to the cyclone. The cyclone returns coarse underflow to the mill and sends fine overflow out as product. The manipulated variables are therefore the ore feed rate MFS, the cyclone feed flow CFF, the sump dilution water SFW, and the mill speed ϕ_c . The controlled variables are the mill charge filling J_T , the sump slurry volume SVOL, and the product particle-size estimate PSE (the fraction of overflow finer than a target mesh). Mill power P_{mill} is monitored as a load and overload indicator. Throughput TP (the product mass rate, approximately MFS at steady state) and PSE are the circuit's economic outputs.

The circuit runs under disturbances on distinct timescales. Ore hardness (f) drifts hour-to-shift and is the dominant source of model-plant mismatch. Feed size and the feed rocks fraction (r) vary over minutes to hours. Liner wear and ball-charge drift act more slowly. The economically meaningful quantity is energy per tonne of product at acceptable quality. The control objective is therefore economic: grind to the required size at the demanded throughput for the least specific energy, while holding the mill inside its safe operating envelope.

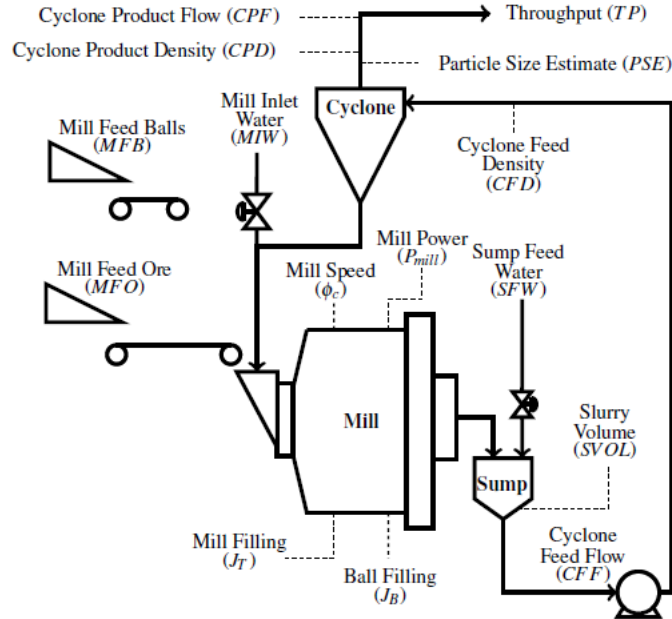


Figure 1. The single-stage SAG grinding mill circuit (after le Roux et al., 2013): SAG mill, sump, and cyclone, with the manipulated variables (MFS, CFF, SFW, c), the controlled variables (JT, SVOL, PSE, Pmill), and the circuit outputs (throughput TP, cyclone-feed density CFD, cyclone product flow CPF and density CPD).

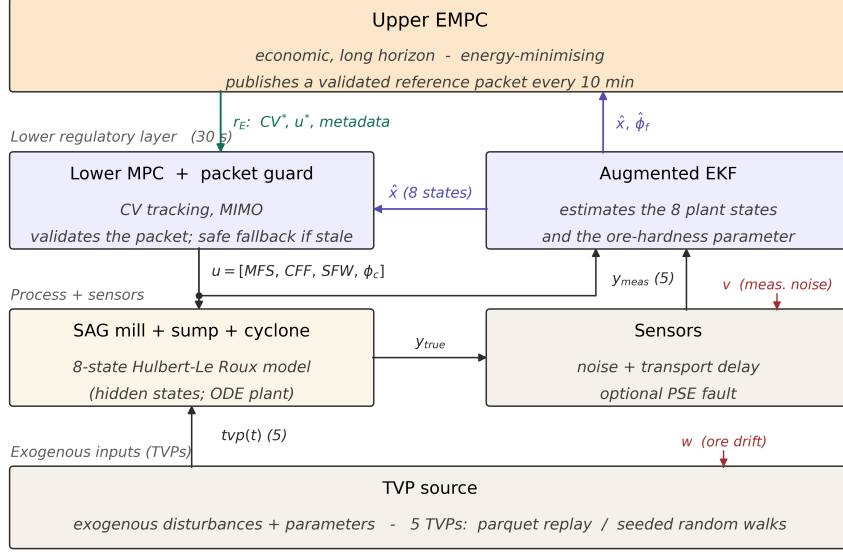
6.2 Control architecture

The controller follows the plant-wide-control hierarchy: a fast regulatory clock and a slower economic clock. Figure 2 shows the two-layer EMPC realisation. At the regulatory clock, a fast non-linear disturbance observer cancels input-side disturbances and a lower MIMO MPC tracks references. At the economic clock, an upper EMPC computes the economically-optimal operating policy over a long horizon, with all four MVs as decision variables. The upper layer does not drive the plant directly. It publishes a reference packet, the CV and MV trajectories $\mathbf{CV}^*$ and $\mathbf{u}^*$. A packet guard validates each packet on solver status, initial-state residual, and age before the lower MPC tracks it. When a solve is rejected, the guard falls back to the last valid packet or a safe set-point. An augmented extended Kalman filter estimates the eight mill and sump states together with the slow ore-hardness parameter $\mathbf{f}$, and feeds both layers.

The economic objective enters the upper EMPC as the modular, mill-local stage cost of Objective 2. The online cost minimises a composable mill-owned term (mill energy, specific energy consumption, or another term from the library of Section 3) subject to the offline-derived particle-size floor. The downstream economics stay confined to the swappable specification interface. Two model families back the prediction step, used in parallel according to the strengths and drawbacks of each (Section 2.2). The Hulbert-Le Roux reduced-order model serves fast online solves. The cumulative-rates PBM serves the cases where the cost must resolve the product size distribution.

Two-layer EMPC control architecture (SAG mill circuit)

Upper EMPC publishes a validated reference packet; the lower MPC tracks it; an augmented EKF feeds both layers.
Upper economic layer (10 min, 4 h horizon)



Upper EMPC (economic, long horizon)

$\min_{x, u} \sum_{k=0}^{N_E-1} \Delta_E B_{NUM} P_{mill}(x_k, u_k, \hat{\phi}_f) + \Delta u \text{ reg} + \text{slacks}$
 $B_{NUM} = 0.06 \text{ \$/kWh}; x_0 = \hat{x}_{now}, x_{k+1} = F_{RK4}(x_k, u_k, \hat{\phi}_f)$
 decision: $x_k(8), u_k(4: MFS, CFF, SFW, \phi_c), \text{slacks}$
 soft zones: $MFS \geq 55, PSE \geq 0.65, JTE \in [0.30, 0.42], SVOLE \in [4.0, 7.5]$
 power cap $P_{mill} \leq 1400 \text{ kW}$; hard MV box (Wakefield bounds)
 publishes r_E every K_E : $CV^* = (JTE^*, SVOL^*, PSE^*), u^*(4), (CV_{ss}, u_{ss})$
 + metadata: IPOPT status, $\|x_E^*(0) - \hat{x}\|$, age; interp. to 30 s grid

Lower MPC (CV tracking) + packet guard

guard: reject r_E if status $\neq$ ok, or $\|x_E^*(0) - \hat{x}\| > \varepsilon$, or age $> 2K_E$
 fallback: hold last-good r_E / safe set-points (defined every tick)
 $\min_u \sum_{k=0}^{N_L-1} \|y_k - CV^*[k]\|_{Q_y}^2 + \|u_k - u^*[k]\|_{R_u}^2 + \Delta u \text{ reg}$
 tracks 3 CVs (not the 8-state x^*); u^* pins ϕ_c ; holds (CV_{ss}, u_{ss})
 back-off bands: $JTE \in [0.306, 0.414], SVOLE \in [4.175, 7.325], MFS \geq 56$
 $N_L = 30, t_s = 30 \text{ s}$; init each tick $x_0 \leftarrow \hat{x}$

— signal (u, y, tv_p) — reference packet r_E — estimator feedback $\hat{x}, \hat{\phi}_f$ — disturbance / noise

Figure 2. The two-layer EMPC control architecture: the upper EMPC (economic, long horizon) publishes a validated reference packet (CV^*, u^*) through a packet guard to the lower CV-tracking MIMO MPC; an augmented EKF estimates the mill and sump states plus the ore-hardness parameter, and a disturbance/parameter (TVP) source drives the simulated plant.

6.3 Forward work plan

The remaining programme is organised into four phases mapped to the objectives of Section 4. Each phase states its method, deliverable, and success criterion.

Phase A - Mill-model calibration and transient-scenario construction (Objectives 1 and 3). *Method.* Obtain historical operating data for a SAG circuit under a data-sharing agreement (Section 8). Fit the Hulbert-Le Roux and cumulative-rates parameters to it. Distinguish identifiable parameters from those fixed by survey data, and validate against held-out operating periods. From the same data, construct a plant-representative input sequence. It combines ore-hardness drift and feed variability (calibrated in amplitude and timescale) with operating-point changes that drive the plant between steady states, where the supervisory architectures are expected to separate. Characterise it by the fraction of runtime spent in transient versus operational steady state. *Deliverable.* A plant-calibrated mill model with a documented identifiability analysis, together with a reproducible transient scenario and its documented statistics. *Success criterion.* Open-loop prediction error within an agreed tolerance on validation data across the normal operating range. The scenario reproduces the disturbance amplitude and timescale statistics of the historical data, with a quantified, non-trivial transient fraction.

Phase B - Modular energy-minimising cost-function formulation (Objective 2). *Method.* Formalise the specification interface. Compile the downstream economics offline into a per-tonne return A and a one-sided PSE/PSD floor derived from downstream recovery and grade behaviour. Define the on-line stage cost as a composable, mill-local objective (energy, SEC, or other mill-owned terms) subject to that floor. Explore cost-function forms for both single- and multi-stage SAG circuits. Each mill optimises its own energy through the interface, so the same cost chains from one mill to the next. Each mill's product-size floor is set by the following mill's feed requirement. *Deliverable.* A modular cost-function specification and its `do-mpc` implementation, with the offline-derivation procedure documented. *Success criterion (the modularity claim, made testable).* Running the same controller across at least two distinct specification interfaces (two ore bodies or two downstream topologies) requires changing only the offline-derived interface. No controller code or structure changes, and the expected operating-point shift appears.

Phase C - Architecture benchmark under the transient scenario (Objective 3). *Method.* Run the full controller set on the Phase A scenario: decentralised PI, tracking MPC, single- and two-layer EMPC, SSRTTO, and ROPA. Hold the estimator stack and the disturbance/sensor model fixed, so that only the supervisory architecture varies. Evaluate on specific energy, constraint compliance (PSE-floor violations), actuation smoothness, and the economic objective, with confidence intervals across scenario realisations. *Deliverable.* A benchmark quantifying the specific-energy gap between architectures under transients.

Success criterion (stated as an a-priori hypothesis). Under transient-dominated operation, the specific-energy and constraint-compliance gaps between gated-idle SSRTO, continuously-adapting ROPA, and dynamic-horizon EMPC widen relative to a single steady-state operating point. The magnitude is quantified with uncertainty.

Phase D - PINN surrogate for online tractability (Objective 4).

Method. Train a physics-informed neural-network surrogate offline on PBM-simulated data, embedding the governing residual in the loss. Substitute it for the mechanistic prediction model inside the EMPC. Quantify the online speed-up and any accuracy or economic cost. *Deliverable.* A PINN surrogate and a tractability/accuracy assessment within the EMPC loop. *Success criterion.* The surrogate reduces online solve time materially. Closed-loop specific energy and constraint compliance stay within an agreed tolerance of the mechanistic-model controller.

7. Project Timeline

The candidature runs over three and a half years, from commencement on 1 March 2026 to thesis submission and defence in September 2029. The Gantt chart below resolves the phases of Section 6.3 to the month. It places the three candidacy milestones: the candidacy report at month 6, the mid-candidacy review at month 18, and the pre-submission review at month 39, three months before the defence. The first six months, now complete, cover the preliminary implementation of the integrated control stack and the Objective 1 baseline reported here. The programme from Phase A onward is the forward plan.

Task	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature review	2026												
Integrated control stack (preliminary)													
Milestone 1: Candidacy (this report)										◆			
Phase A: calibration + scenario (O1, O3)													
Phase A: calibration + scenario (O1, O3)	2027												
Phase B: modular cost function (O2)													
Milestone 2: Mid-Candidacy (Month 18)										◆			
Phase C: architecture benchmark (O3)													
Phase C: architecture benchmark (O3)	2028												
Phase D: PINN surrogate (O4)													
PINN-EMPC integration and tractability (O4)													
Industrial implementation and interfacing													
Industrial implementation and interfacing	2029												
Industrial validation and integration													
Analysis, reporting and dissemination													
Thesis writing													
Milestone 3: Pre-Submission													
Thesis submission and defence													

8. Ethics, Permits and Clearances

This research does not involve human or animal participants, so formal ethics clearance is not required. The work is computational: simulation, control design, and data analysis. One non-trivial requirement does apply. The plant-calibration and transient-scenario phase (Phase A, Section 6.3) depends on historical operating data from an industrial SAG circuit. That will require a data-sharing agreement and, where applicable, a non-disclosure agreement with the providing operator. Confidential data will be handled, stored, and reported in accordance with the agreement terms and the Australian Code for the Responsible Conduct of Research. Until such data is secured, the methodology can be exercised on a synthetic plant that mimics the calibrated statistics.

9. Data Management

All code, simulation configurations, model parameters, and analysis notebooks are version-controlled in a GitHub repository. The repository already holds the controller implementations, the literature review, and the implementation walkthrough. Development and data analysis are carried out on a Curtin-managed laptop. The codebase and the generated artefacts (simulation outputs, figures, trained model/estimator parameters) are backed up to the Curtin OneDrive area. Any confidential industrial data will be held separately under access control per the data-sharing agreement. It will be stored encrypted, decrypted only during data processing, and never committed to the repository. Research data will be retained for a minimum of seven years in compliance with the Australian Code for the Responsible Conduct of Research.

10. Resources and Budget

The work is primarily computational. The core toolchain is `do-mpc`/CasADi with the IPOPT NLP solver, plus MATLAB/Simulink for cross-validation. All of it is available under Curtin licensing or as open-source software. It runs on a workstation with adequate CPU, and a GPU for the PINN phase. Cloud compute is a contingency for the larger sweep and training runs. Indicative budget items follow the standard HDR categories.

Item	Indicative cost	Source
Workstation / compute upgrade (CPU + GPU for PINN)	[amount]	HDR consumables
Cloud compute (contingency, sweeps + PINN training)	[amount]	HDR consumables

Item	Indicative cost	Source
Software licences (MATLAB toolboxes; remainder open-source)	[amount]	Faculty / HDR
Total	[total]	

[Populate amounts to match the Curtin HDR budget template and your funding sources.]

11. Conclusion

Comminution dominates a concentrator’s electricity bill. A SAG circuit spends most of its life moving between operating points, not holding the strict steady state that classical real-time optimisation assumes. This report has argued that the natural control objective is therefore economic. At the SAG-mill level it reduces to minimising specific energy at the demanded throughput and quality. Preliminary work has implemented the full controller spectrum (tracking MPC, single- and two-layer EMPC, SSRTO, and ROPA) on a simulated Hulbert-Le Roux circuit. All of them behave as theory predicts. The work also exposed the study’s central observation. At a single operating point, the economic and RTO architectures are nearly indistinguishable. Any honest comparison must therefore be run under plant-representative transients. The candidature’s two core contributions follow directly. The first is a modular, energy-minimising cost function that confines the downstream economics to a swappable offline interface. The second is a transient-rich benchmark that quantifies where the architectures genuinely diverge, supported by an integrated disturbance-observer plus augmented-EKF estimator stack. The remaining programme calibrates the model to plant data, constructs and characterises the transient scenario, and delivers the comparison. The higher-risk PINN objective is staged so the thesis remains complete regardless of its outcome. The end goal is practical. The modular energy-minimising cost function should run inside an EMPC/MPC controller that is validated and integrated against industrial data. The work then delivers a controller an operating concentrator can deploy, and not only a simulation result.

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